CROSSSAFE · PROBLEM STATEMENT

CrossSafe problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-31	Amish Chadha	Kit 1.9.1 upgrade: users and needs table and stakeholders table added (source of CRS-URS-001)
0.5	2026-10-02	Amish Chadha	Pilot partner open question answered with the selection rule and first candidate decided on 2026-10-02
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.3	2026-09-25	Amish Chadha	TRL 3 update; budget scope per CRS-DDR-001 D1, IA-21 details checked, open questions linked to CRS-DDR-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, out of scope, prior work)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CrossSafe problem statement

Pedestrians die at unsignalized crossings, and full traffic signals are costly and slow to approve. A driver who does not see a waiting pedestrian, or sees them too late, does not yield. CrossSafe is an open, solar-powered beacon that warns drivers only when someone is actually waiting to cross, at crossings near schools, markets and bus stops that will not get a full signal. Design with, not for: requirements must come from co-design sessions with road users and the road authority through a local partner.

The problem

About 1.19 million people die on the world's roads each year, and pedestrian deaths rose 3 % to about 274,000 between 2010 and 2021, 23 % of the total (WHO, 2023). Nine in ten road deaths occur in low- and middle-income countries (WHO, 2023), and road traffic injuries are the leading cause of death for children and young people aged 5 to 29 (WHO fact sheet).

High-income countries are not exempt. In the United States, 7,314 pedestrians were killed in 2023; 74 % of those deaths happened away from intersections and 77 % in the dark (NHTSA, 2025). In Great Britain, pedestrians were 25 % of the 1,624 road deaths in 2023 (Department for Transport).

The marked crossing without a signal is where the pedestrian's right of way and the driver's attention most often fail to meet. Three gaps make it hard to fix:

1. **Cost and approval.** A full signal needs a mains connection, a cabinet, foundations and an engineering study. Many crossings will never qualify or be funded.
2. **Static signs fade into the background.** A sign that is always there says nothing about whether someone is waiting now. Warning lights that flash all day teach drivers to ignore them.
3. **Power and maintenance.** Many crossings that need help are on roads without a convenient mains supply, and closed commercial beacons are expensive to buy and hard to repair locally.

There is good evidence that an active, pedestrian-triggered warning works. The US Federal Highway Administration lists the rectangular rapid flashing beacon (RRFB) as a proven safety countermeasure that can reduce pedestrian crashes by up to 47 % and raise driver yielding rates to as high as 98 %, depending on speed, lanes, crossing distance and time of day (FHWA). CrossSafe asks whether an open, solar, garage-buildable design can bring that kind of device to crossings and communities that cannot buy one today.

Users and context

User	Need	Context
Pedestrian, including children, older people and people with disabilities	Cross safely, with drivers stopping; clear confirmation that the warning is on	School run, market days, bus stops, often at dusk or in the dark
Driver	Early, unmistakable warning only when someone is waiting	Urban and peri-urban roads, 30 to 60 km/h, glare and rain
School or market community	A safer crossing without waiting years for a signal	Parent groups, school boards, market associations as local champions
Road authority or municipal engineer	A device that meets local rules, is reliable, and costs little to maintain	Approval, installation permits, asset registers
Local technician	Install, service and repair with ordinary tools and parts	Hand tools, a ladder or bucket truck, a multimeter

Users and needs

Table 1a. Users and needs (source of the user requirements in CRS-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Pedestrian, including children, older people and people with disabilities	Cross with drivers stopping; know the warning is on; reach and press the button	Waiting at a marked crossing on the school run, at a market or a bus stop, often at dusk	Drivers who do not yield; a button that is out of reach or silent; not knowing if the call was registered
Driver	Early, unmistakable warning only when someone is waiting	Driving 30 to 60 km/h on an urban or peri-urban road in glare or rain	Beacons that flash all day and are ignored; warnings seen too late to stop
School or market community	A safer crossing without years of waiting; low cost; works without mains	Parent groups and market associations raising money and finding a site	Cost and approval of a full signal; theft and vandalism; nobody to maintain it
Road authority or municipal engineer	A device that meets local rules, is reliable and cheap to maintain	Approval, installation permit, asset register, yearly inspection	Closed products that cannot be repaired; unproven devices; liability
Local technician	Make, install and repair with ordinary tools and parts	A workshop or a municipal crew with hand tools, a ladder and a multimeter	Special parts, no documents, a battery in a hot box

Stakeholders

Table 1b. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Road authority (traffic engineering department)	Approves any use on a public road; sets colours, flash pattern, sign type and mounting height	Pilot partner selection rule decided 2026-10-02; early review of the design against local rules (TRL 4)
School boards, parent groups, market associations	Local champions who fund and host a crossing	Co-design sessions on the button, the warning time and the site (TRL 4)
Disability and older-adult groups	Accessibility of the button, the sound, the tactile arrow and the light	Review of PR-011 and the button station in co-design
Workshops and makerspaces	Build the made parts and service the device	Review of the build plan CRS-BLD-001 for local skills and tools
Utility owner of an existing pole	Owns poles the kit may clamp to	Permission and pole condition check per site, to assess
Amish Chadha (Design Molecule)	Decides scope, budget, pitch and safety trade-offs	Decisions listed in CRS-DEC-001

Operating environment

- **Site:** a marked, unsignalized crossing on a two-lane road, about 7 m kerb to kerb, with a sidewalk or verge on each side. Wider roads need a median assembly (out of scope for the first concept).
- **Climate:** ambient -20 to +50 °C across target regions, direct sun on the enclosure, heavy rain, dust and road spray.
- **Power:** no mains assumed; solar resource from about 2.5 peak sun hours per day in a poor month (estimate for mid latitudes and tropical rainy seasons) upward.
- **Use:** up to about 300 activations per day at a busy school or market crossing (estimate, to be checked in co-design and site counts).
- **Threats:** theft of panels and batteries, vandalism, vehicle strikes on the pole.

Constraints

- Garage-buildable prototype of one beacon assembly on an existing pole, a value-engineering target of $350 USD in parts (`project.yaml` ; scope adopted for TRL 3 in CRS-DDR-001 D1, open for Amish's review). A crossing needs two assemblies; a value-engineering target of $750 applies to a two-sided site trial when one starts (CRS-DDR-002, O2).
- Built from off-the-shelf parts: standard signs, LED modules, a small solar kit, a LiFePO4 battery, an IP65 box and a common microcontroller.
- Needs road authority approval before any use on a public road. Local rules on colors, flash patterns, sign design and mounting heights take precedence over anything in this repo.
- Privacy: counts and activation events only; no images or audio are recorded or leave the device.

Out of scope

- Full traffic signals, pedestrian hybrid beacons and any device that shows a red indication or stops traffic.
- Enforcement features such as speed cameras.

- Crossings of more than one lane per direction, where a beacon alone is not enough and a median refuge or signal is usually needed.
- Certification to any national standard; the repo can only document how the design relates to one.

Prior work

- **RRFB.** Amber, rectangular, pedestrian-activated beacons mounted with the crossing warning sign. In the United States they are covered by FHWA Interim Approval IA-21 (20 March 2018), which sets the minimum indication size (at least 5 in wide by 2 in high), daytime intensity to the SAE J595 Class 1 yellow peak, a wig-wag flash pattern at 75 flashing sequences per minute, automatic night dimming, pushbutton or passive detection, an optional pilot light for pedestrians, and use only at uncontrolled marked crosswalks (FHWA IA-21). Solar RRFBs are sold commercially; they are closed designs.
- **Evidence of effect.** FHWA's proven safety countermeasure summary for RRFBs (see above) gives the crash and yielding figures used in this repo.
- **Sibling designs in the lab.** FieldNode (a sibling Design Molecule repo) is the lab's shared solar sensor node (6 W panel, 3.2 V LiFePO4 cell, LoRaWAN). Its radio and logging core (an STM32WL-class module) is reused for the CrossSafe controller, but its power stage is too small for the LED load; see CRS-PRC-001 and CRS-DDR-001 D7.

Open questions

- [ ] Which pilot country and road authority, and which rules apply there (colors, flash pattern, sign type, mounting height)? Decided 2026-10-02 as a selection rule: a United States city traffic engineering department with a Safe Routes to School or Vision Zero programme; first candidate to approach, a North Texas city near the team (CRS-DEC-001).
- [ ] Is passive detection allowed and trusted locally, or should the button be the only trigger?
- [ ] How many activations per day at the candidate sites, and how dark are they at the busy times?
- [ ] How much theft and vandalism should the design expect, and at what mounting height is the battery safe?

User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [ ] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate load, distance, terrain and cost assumptions in the field
- [ ] Revise requirements (REQ) from findings before freezing the design